

# LAB 0 :

## 1.What is SQL and its Use?

- SQL stands for **Structured Query Language**.
- SQL is a standard language used for **accessing and manipulating data in databases**.
- SQL is primarily used with **Relational Database Management Systems (RDBMS)**.
- SQL allows users to **create, modify, retrieve, and manage data** in a database.
- It is used to **create and manage database objects**, such as tables and views.
- SQL also provides commands for **controlling user access and managing transactions**.
- SQL acts as a communication interface between **software applications and database management systems**.

## 2. Describe all the different commands of SQL, mention use of it and syntax.

### 1.DDL - Data Definition Language

#### 1. CREATE

- USE - create database or its objects.
- SYNTAX -
- CREATE TABLE

Table\_name(column1 data\_type , column2 data\_type);

#### 2. DROP

- USE - delete objects from the database.
- SYNTAX - DROP TABLE table\_name;

#### 3. ALTER

- USE - alter the structure of the database.
- SYNTAX - ALTER TABLE table\_name ADD COLUMN column\_name data\_type;

#### 4. TRUNCATE

- USE - Remove all the records from a table, including all spaces allocated for the records are removed.
- SYNTAX - TRUNCATE TABLE table\_name;

#### 5. COMMENT

- USE - Add comments to the data dictionary.

#### 6. RENAME

- USE - Rename an object existing in the database.
- SYNTAX - RENAME TABLE old\_table\_name TO new\_table\_name;

## 2. DQL - DATA QUERY LANGUAGE

### 1. SELECT

- USE - It is used to retrieve data from the database
- SYNTAX - SELECT column1, column2, ... FROM table\_name WHERE condition;

### 2. FROM

- USE - Indicates the table from which to retrieve the data.
- SYNTAX - SELECT column1 FROM table\_name;

### 3. WHERE

- USE - Filters rows before any grouping or aggregation
- SYNTAX -

SELECT column1

FROM table\_name

WHERE condition;

### 4. GROUP BY

- USE - Groups rows that have the same values in specified columns.

- SYNTAX- SELECT column1, AVG(column2)

FROM table\_name

GROUP BY column1;

### **3. DML - Data Manipulation Language**

#### **1. INSERT**

- USE-Insert data into a table
- SYNTAX-INSERT INTO table\_name (column1, column2, ...) VALUES (value1, value2, ...);

#### **2. UPDATE**

- USE -Update existing data within a table
- SYNTAX-UPDATE table\_name SET column1 = value1, column2 = value2 WHERE condition;

#### **3. DELETE**

- USE - Delete records from a database table
- SYNTAX-DELETE FROM table\_name WHERE condition;

### **4. DCL - Data Control Language**

#### **1. GRANT**

- USE -Assigns new privileges to a user account, allowing access to specific database objects, actions or functions
- SYNTAX-GRANT privilege\_type [(column\_list)] ON [object\_type] object\_name TO user [WITH GRANT OPTION];

#### **2.REVOKE**

- USE - Removes previously granted privileges from a user account, taking away their access to certain database objects or actions
- SYNTAX-REVOKE [GRANT OPTION FOR] privilege\_type [(column\_list)] ON [object\_type] object\_name FROM user [CASCADE];

## 5. TCL - Transaction Control Language

### 1. BEGIN TRANSACTION

- USE -Starts a new transaction
- SYNTAX-BEGIN TRANSACTION [transaction\_name];

### 2. COMMIT

- USE- Saves all changes made during the transaction
- SYNTAX-COMMIT;

### 3. ROLLBACK

- USE- Undoes all changes made during the transaction
- SYNTAX- ROLLBACK;

### 4. SAVEPOINT

- USE- Creates a savepoint within the current transaction
- SYNTAX-SAVEPOINT savepoint\_name;

# 1. Write a step by step procedure to install MYSQL.

## Step 1: Visit the Official MySQL Website

Open your preferred web browser and navigate to the official MySQL website. Download MySQL Installer for Windows from the official MySQL website.

**General Availability (GA) Releases** Archives

### MySQL Installer 8.0.35

**Note:** MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:  
8.0.35

Select Operating System:  
Microsoft Windows

|                                                                                             |        |        |                          |
|---------------------------------------------------------------------------------------------|--------|--------|--------------------------|
| <b>Windows (x86, 32-bit), MSI Installer</b><br>(mysql-installer-web-community-8.0.35.0.msi) | 8.0.35 | 2.1M   | <a href="#">Download</a> |
| MD5: 214df2ccdf83eb5edc6ca7c115792406   Signature                                           |        |        |                          |
| <b>Windows (x86, 32-bit), MSI Installer</b><br>(mysql-installer-community-8.0.35.0.msi)     | 8.0.35 | 288.6M | <a href="#">Download</a> |
| MD5: 2cfda448a2971b6b5323775ef9e8d012   Signature                                           |        |        |                          |

## Step 2: Go to the Downloads Section

On the MySQL homepage, click on the "No thanks, just start my download" link to proceed to MySQL downloading.

## MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

**Login »**

using my Oracle Web account

**Sign Up »**

for an Oracle Web account

MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can sign up for a free account by clicking the Sign Up link and following the instructions.

**No thanks, just start my download.**

### Step 3: Run the Installer

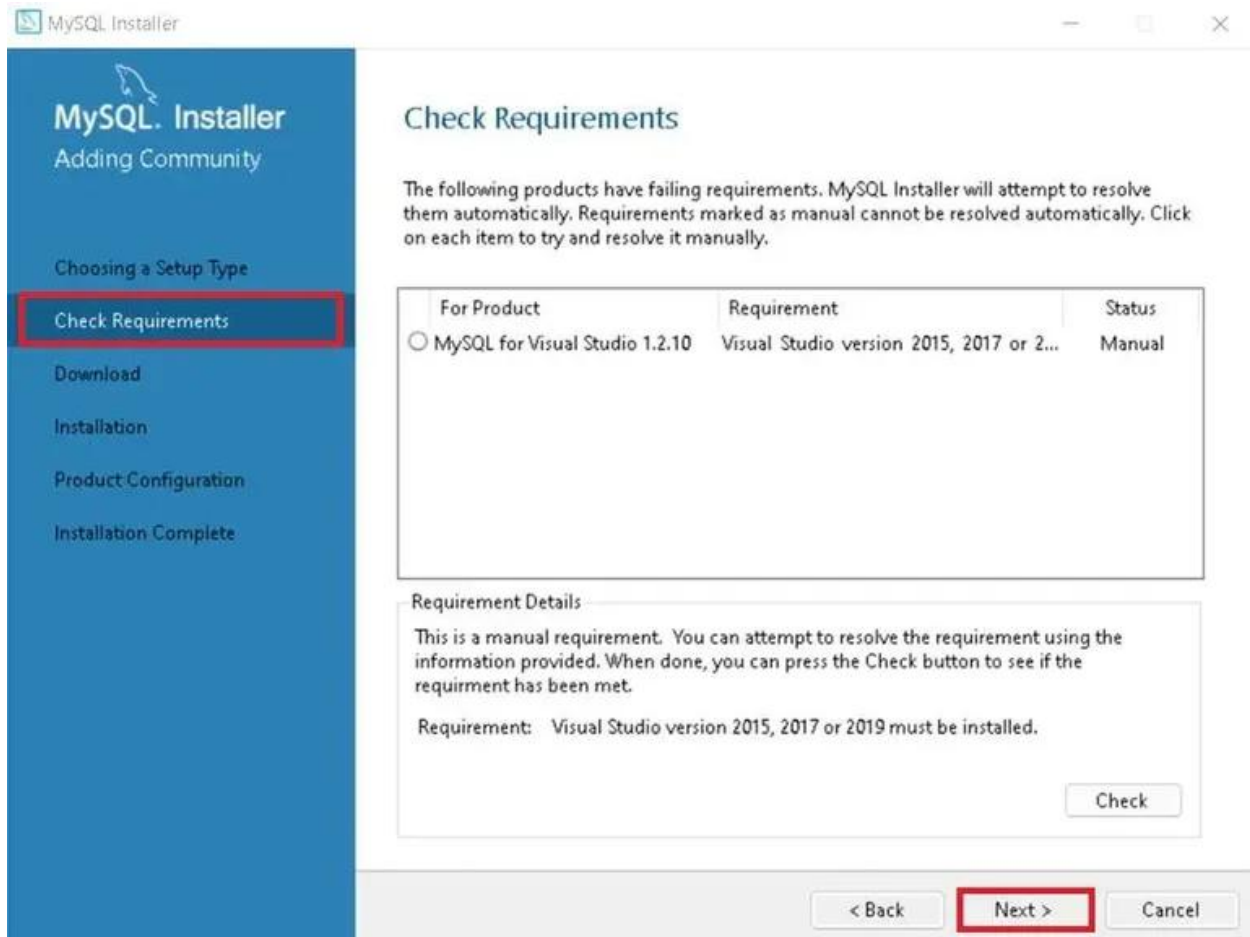
After downloading the MySQL installer (.exe file), go to your Downloads folder, find the file and double-click to run the installer

### Step 4: Choose Setup Type

The installer will instruct you to choose the setup type. For most users, the "Developer Default" is suitable. It includes MySQL Server, MySQL Workbench, and other development tools. Click "Next" to proceed.

## Step 5: Check Requirements

The installer will check for required dependencies such as the Microsoft Visual C++ Redistributable. It may automatically resolve missing components, but in some cases, manual installation may be required.



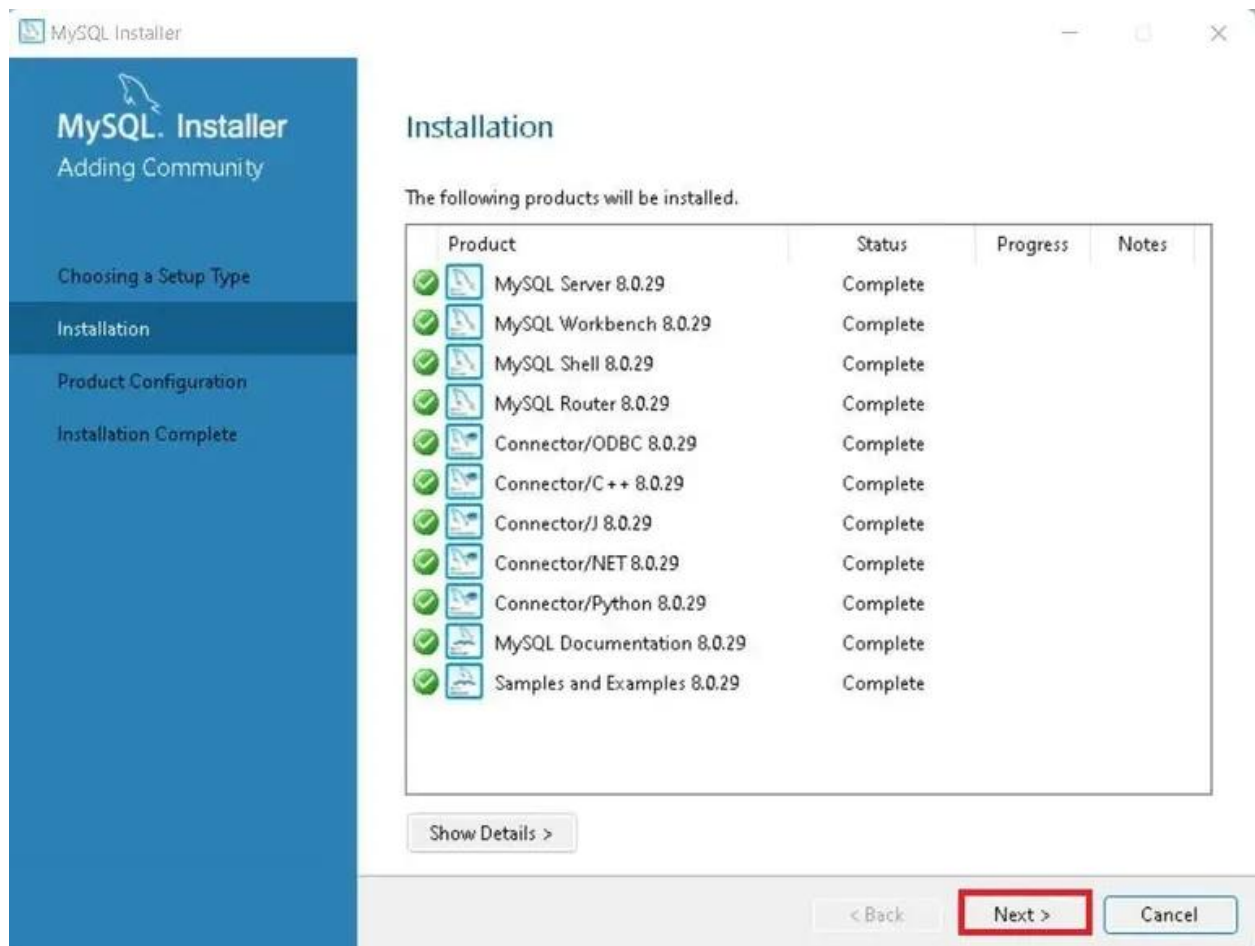
## Step 6: MySQL Downloading

Now that you're in the download section, click "Execute" to start downloading the components you selected. . MySQL Server, MySQL Workbench, and other required components will be downloaded .Wait a

few minutes until all items show tick marks, indicating completion, before moving forward.

## Step 7: MySQL Installation

Now the downloaded components will be installed. Click "Execute" to start the installation process. MySQL will be installed on your Windows system. Then click Next to proceed.



## Step 8: Navigate to Few Configuration Pages

Proceed to "Product Configuration" > "Type and Networking" > "Authentication Method" Pages by clicking the "Next" button.

## Step 9: Create MySQL Accounts

Create a password for the MySQL root user. Ensure it's strong and memorable. Click "Next" to proceed.

The screenshot shows the MySQL Installer window for MySQL Server 8.0.29. The left sidebar contains the following steps: Type and Networking, Authentication Method, Accounts and Roles (highlighted), Windows Service, and Apply Configuration. The main content area is titled 'Accounts and Roles' and contains two sections. The first section, 'Root Account Password', prompts the user to enter a password for the root account, with fields for 'MySQL Root Password' and 'Repeat Password', and a 'Password strength' indicator showing 'Strong'. The second section, 'MySQL User Accounts', prompts the user to create MySQL user accounts and assign roles, with a table for 'MySQL User Name', 'Host', and 'User Role', and buttons for 'Add User', 'Edit User', and 'Delete'. At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a red box), and 'Cancel'.

MySQL Installer

MySQL Server 8.0.29

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

### Accounts and Roles

Root Account Password

Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password strength: Strong

### MySQL User Accounts

Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

| MySQL User Name | Host | User Role |
|-----------------|------|-----------|
|-----------------|------|-----------|

Add User

Edit User

Delete

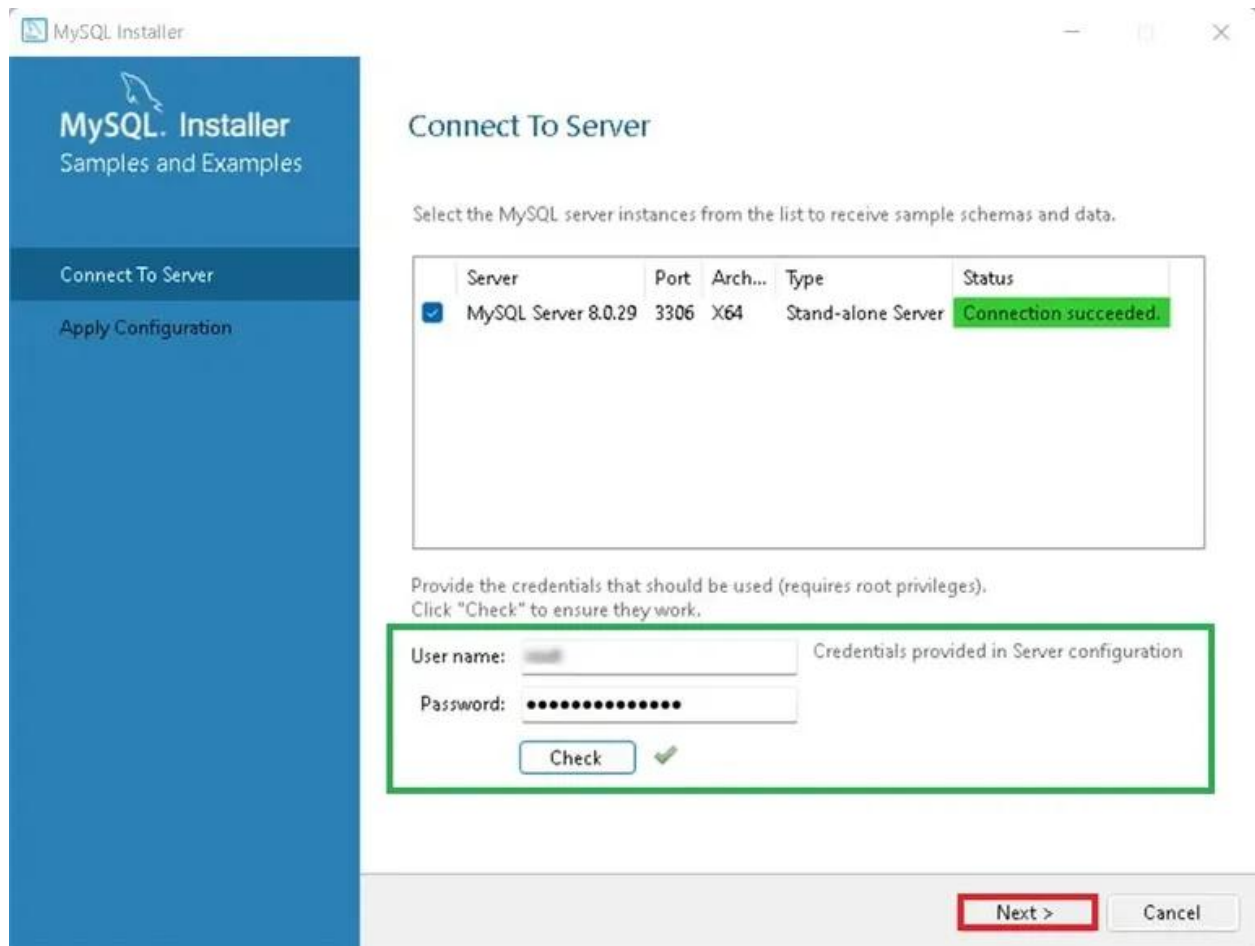
< Back

Next >

Cancel

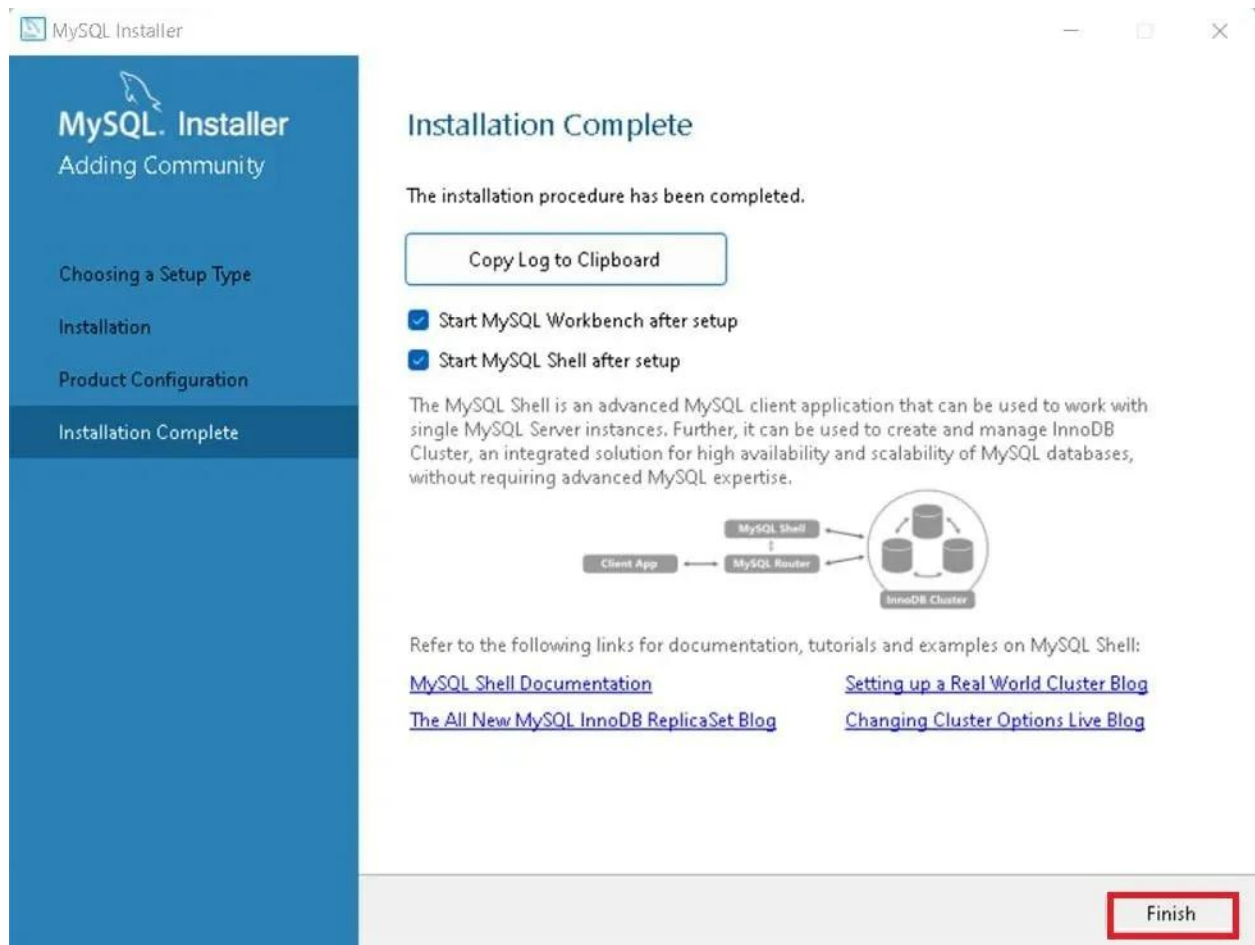
## Step 10: Connect To Server

Enter the root password, click Check. If it says, "Connection succeed," you have successfully connected to the server.



## Step 11: Complete Installation

Once the installation is complete, click "Finish." Congratulations! MySQL is now installed on your Windows system.



## Step 12: Verify Installation

To ensure the successful installation of MySQL, open MySQL Workbench or the MySQL Command Line Client, both available in your Start Menu. Log in using the root user credentials you set during installation. If the connection is successful, MySQL and MySQL Workbench have been installed correctly.

